

Personal Portfolio Website

Case Study — Sample Work

HTML5 • CSS3 • JavaScript (ES6)

Project Overview

This project is a responsive personal portfolio website built from scratch using HTML, CSS, and vanilla JavaScript. It was designed to showcase professional experience, technical skills, and project work in a single-page layout, while giving me hands-on practice writing front-end code without relying on a framework.

The site includes a hero section with a profile introduction, an About Me section, a Work Experience timeline, an Education section, a Projects showcase, a Skills summary, and a Contact section — all navigable from a fixed top navigation bar.

Key Features

- Dark mode toggle with smooth color-scheme transitions across the whole page
- Sticky navigation bar with anchor links to each page section
- Image modal / lightbox — clicking the profile photo opens an enlarged view with a close button and Escape-key support
- Scroll-triggered animations that fade and slide content into view as the user scrolls (hidden /show classes)
- Interactive "Say Hello" button that dynamically updates page text using the DOM API
- Project cards for each major work sample, each with "View Details" and "View PDF" actions
- Fully responsive layout that adapts across desktop and mobile viewports

1. HTML Structure

The markup below shows the semantic structure of index.html: the document head links Font Awesome for icons and a custom style.css for styling, and the body opens with an image modal and a hero section containing the profile picture, name, title, and an introductory call-to-action button.

```
HTML > > index.html > html > body > section#projects > div.projects-container > div.project-card
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <button id="darkModeBtn">🌙 Dark Mode</button>
5
6   <title>Ell's Portfolio</title>
7
8   <!-- Font Awesome Icons -->
9   <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/font-awesome@6.7.2/css/all.min.css">
10
11   <!-- Your CSS -->
12   <link rel="stylesheet" href="style.css">
13 </head>
14 <div id="imageModal" class="modal">
15   <span class="close">&times;</span>
16
17   <img class="modal-content" id="fullImage">
18
19 </div>
20 <body>
21
22 <div class="hero">
23
24   
25
26   <div class="hero-text">
27     <h1>Hi! I'm Ell</h1>
28     <p>Web Developer | Technical Support Engineer | QA Tester </p>
29     <p id="welcomeText">Welcome to my personal portfolio website!</p>
30
31     <button id="welcomeBtn">Say Hello 🙋</button>
32
33   </div>
34
```

Fig. 1 — index.html: document head and hero section markup

2. CSS Styling

The stylesheet defines the site's visual language, including the semi-transparent, blurred navigation bar, the dark-mode toggle button and its hover state, and the scroll-reveal animation classes (hidden and show) that drive the fade-in-on-scroll effect used throughout the page.

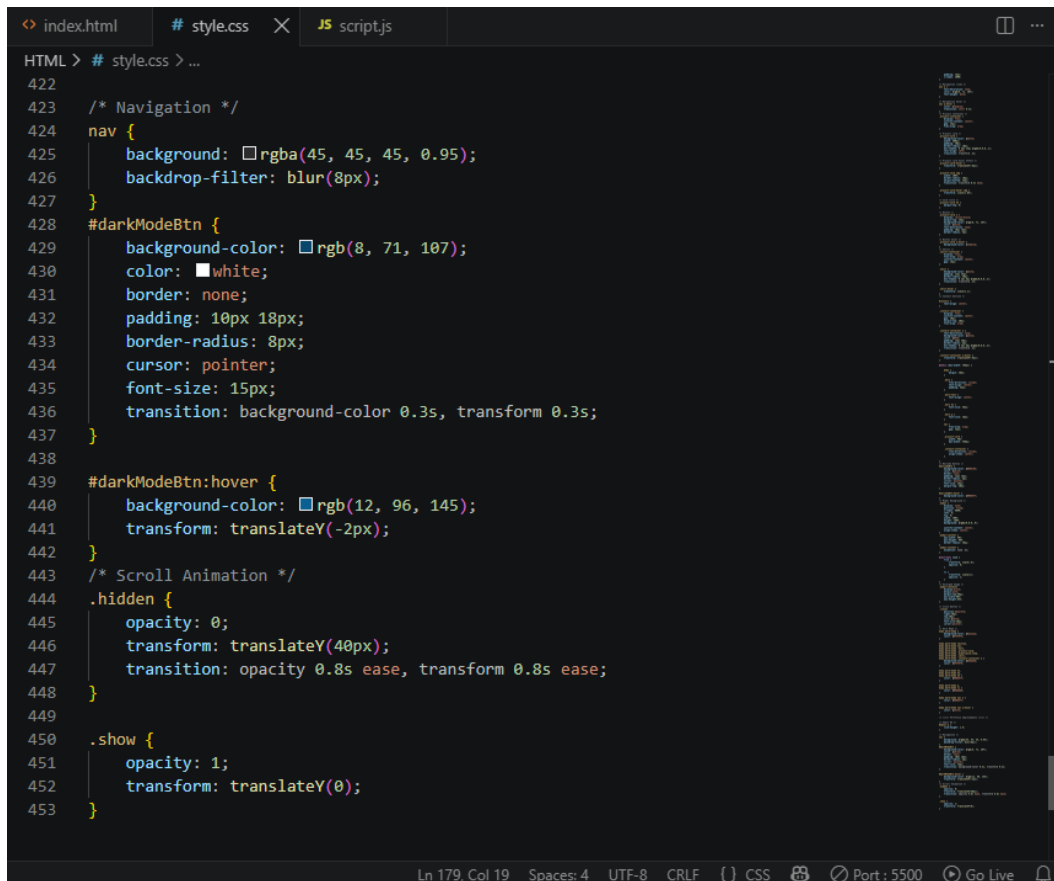
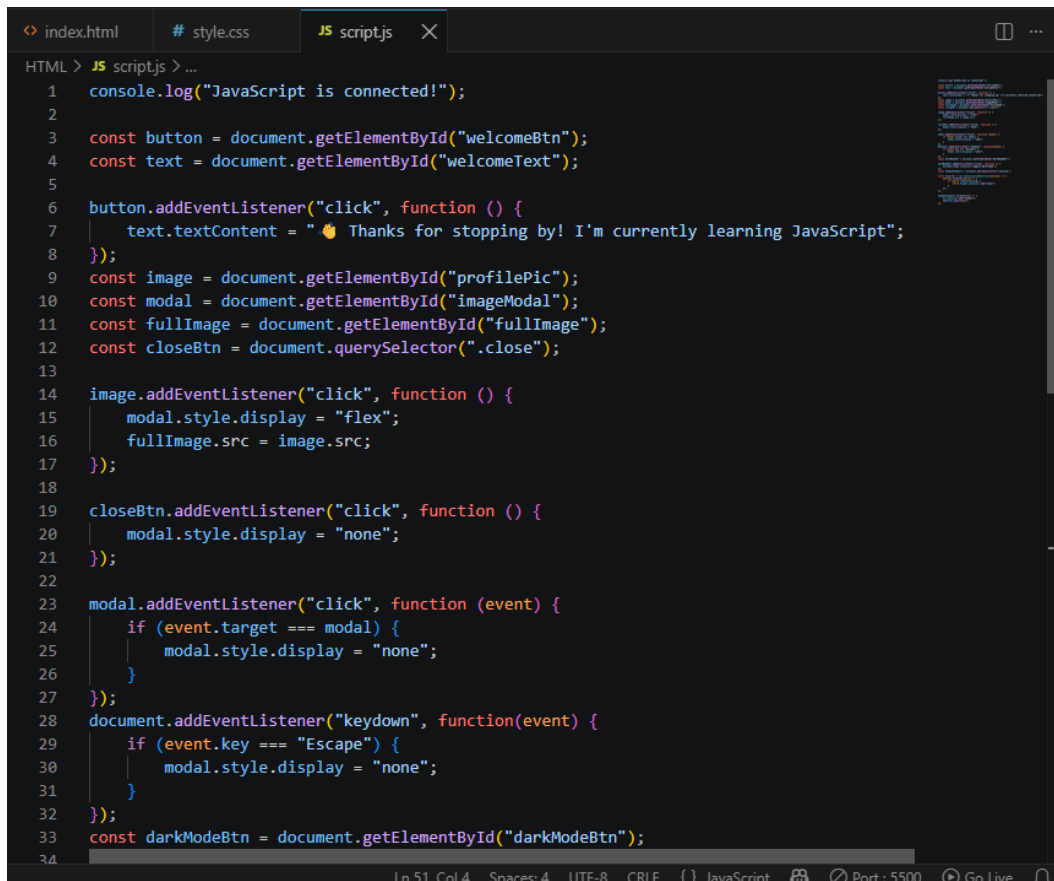
A screenshot of a code editor with a dark theme. The editor has three tabs at the top: 'index.html', '# style.css', and 'JS script.js'. The '# style.css' tab is active, showing CSS code. The code includes a 'nav' rule with background and backdrop-filter properties, a '#darkModeBtn' rule with background-color, color, border, padding, border-radius, cursor, font-size, and transition properties, and a '#darkModeBtn:hover' rule with background-color and transform properties. There are also '.hidden' and '.show' rules for scroll animation. The status bar at the bottom shows 'Ln 179, Col 19', 'Spaces: 4', 'UTF-8', 'CRLF', '() CSS', 'Port: 5500', and 'Go Live'.

Fig. 2 — style.css: navigation, dark-mode button, and scroll-animation rules

3. JavaScript Interactivity

script.js wires up the page's interactive behavior: a click listener on the "Say Hello" button swaps in new welcome text, and a set of listeners power the profile-picture modal — opening it on image click, closing it via the close button, a click outside the modal, or the Escape key.

A screenshot of a code editor with a dark theme. The editor has three tabs at the top: 'index.html', '# style.css', and 'JS script.js'. The 'JS script.js' tab is active. The code is written in JavaScript and handles DOM events for a welcome button and an image modal. The code includes comments and uses modern JavaScript syntax like 'const' and 'addEventListener'. The editor's status bar at the bottom shows 'Ln 51, Col 4', 'Spaces: 4', 'UTF-8', 'CRLF', and 'JavaScript'.

```
HTML > JS script.js > ...
1 console.log("JavaScript is connected!");
2
3 const button = document.getElementById("welcomeBtn");
4 const text = document.getElementById("welcomeText");
5
6 button.addEventListener("click", function () {
7     text.textContent = "👋 Thanks for stopping by! I'm currently learning JavaScript";
8 });
9 const image = document.getElementById("profilePic");
10 const modal = document.getElementById("imageModal");
11 const fullImage = document.getElementById("fullImage");
12 const closeBtn = document.querySelector(".close");
13
14 image.addEventListener("click", function () {
15     modal.style.display = "flex";
16     fullImage.src = image.src;
17 });
18
19 closeBtn.addEventListener("click", function () {
20     modal.style.display = "none";
21 });
22
23 modal.addEventListener("click", function (event) {
24     if (event.target === modal) {
25         modal.style.display = "none";
26     }
27 });
28 document.addEventListener("keydown", function(event) {
29     if (event.key === "Escape") {
30         modal.style.display = "none";
31     }
32 });
33 const darkModeBtn = document.getElementById("darkModeBtn");
34
```

Fig. 3 — script.js: DOM event handling for the welcome button and image modal

4. Projects & Skills Section

The Projects section presents three work samples as cards — this Personal Portfolio Website, QA Testing Documentation, and Technical Support & Troubleshooting — each with a short description, a technologies-used line, and buttons to view further detail or a PDF write-up. A Skills section below lists the core competencies demonstrated across these projects.

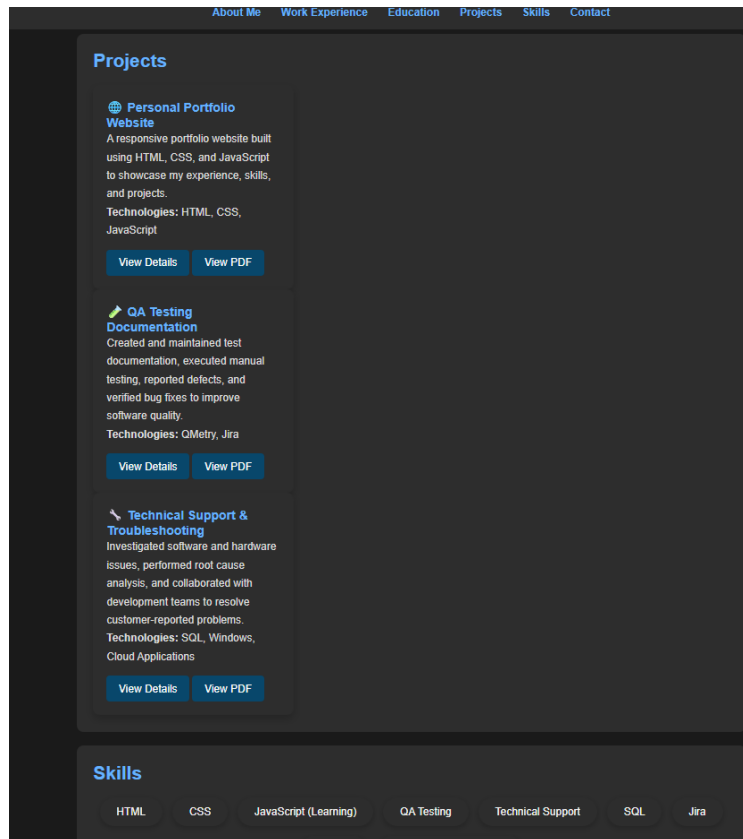


Fig. 4 — Rendered Projects and Skills sections of the live site

5. Live Site — Home View

The finished site as rendered in the browser, showing the hero section with dark mode enabled, the profile photo, title line, and the fixed navigation bar linking to every section of the page.

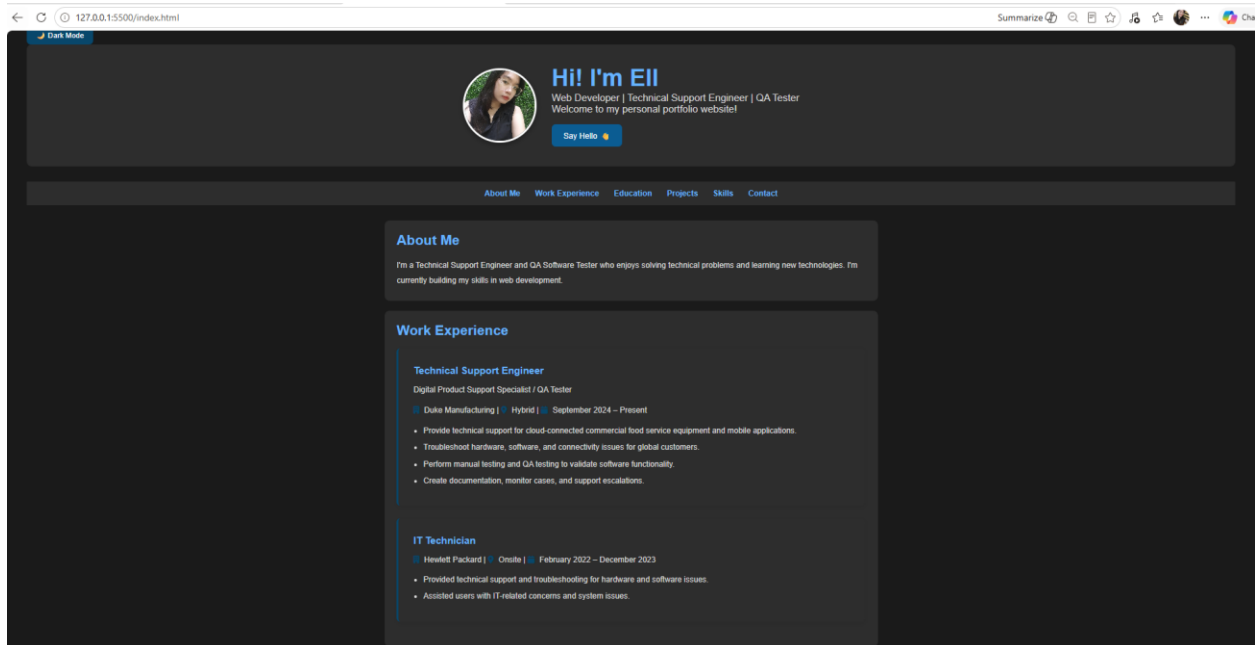


Fig. 5 — Live preview: hero section and navigation bar in dark mode

Technologies Used

- HTML5 — semantic page structure
- CSS3 — responsive layout, transitions, dark-mode theming
- JavaScript (ES6) — DOM manipulation, event handling, modal and scroll interactions

Outcome

The result is a fast, dependency-free portfolio site that loads instantly, works without a build step, and clearly communicates my background as a Web Developer, Technical Support Engineer, and QA Tester. It doubles as a live demonstration of core front-end fundamentals: semantic HTML, custom CSS theming, and vanilla JavaScript interactivity.